Joshua Cox

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Education

University of Illinois Urbana-Champaign
Master of Science, Mechanical Engineering *GPA: 3.83 / 4.00* *Expected: December 2025*
Bachelor of Science, Mechanical Engineering | Minor, Computer Science *GPA: 3.61 / 4.00* *December 2022*

Skills

Programming Languages: Python, C/C++, MATLAB, Java, C#, Microsoft VBA
Software: Simulink, PyTorch, Docker, Git, ROS, STM Studio, Godot, pygame, Unity CARLA Simulation, LaTeX
Hardware: STM32 Nucleo boards, TI C2000 MCU boards, Stretch3, UR3 Robot arm, CRS Robot arm, Arduino, Raspberry Pi

Research Experience

Lunar Autonomy Challenge | Human-Centered Autonomy Lab *November 2024 - Present*
- Founded team to compete in NASA software challenge to autonomously navigate and map lunar surface using a rover
- Implemented navigation strategy employing both a high and low-level planner, creating local object avoidance methods, querying rover's battery usage, and determining heuristics on when to recalculate path
- Created 2D exploration simulation from scratch with python and pygame in order to test team's navigation approach

Handle Haptics | Human-Centered Autonomy Lab *February 2024 - Present*
- Designed, created CAD for, and 3D-printed parts for mounting strategies to attach vibration sensors onto a Hello Robot Stretch3 robot to better communicate directions and signals to guided users
- Rapidly prototyped initial designs including a technique to attach Nintendo Joy-Cons in various positions

Anomaly Detector Informed Learning | Human-Centered Autonomy Lab *August 2023 - Present*
- Composed simulation environment utilizing CARLA to demonstrate data gathering and learning of mobile robots
- Tested if using an anomaly detector improves imitation learning performance on mobile robots by identifying unfamiliar scenarios and prompting an expert to intervene

Work Experience

Control Systems Teaching Assistant | University of Illinois Urbana-Champaign | Urbana, IL *January 2023 – Present*
- Directly TA'd the lab section of numerous engineering classes related to controls, signals, and embedded systems
- Took on a managerial role at times to teach other TAs material and their responsibilities, develop course work, redesign documentation, prepare class resources, and fix equipment as needed

Embedded Systems Engineer Intern | Engineering System Design Lab | Urbana, IL *May 2024 – August 2024*
- Completed hardware and software tasks to develop a new testing setup for a hinge-integrated multifunctional structures for attitude control (MSAC) scheme for satellites utilizing piezoelectric strain actuators
- Programmed software in C++ for STM32 microcontroller Nucleo-H755ZI-Q for the purpose of enabling advanced PWM signals, logging SD card data, and communicating with multiple accelerometers
- Tested types of signal filters to produce sin signals from period-changing PWM signals as inputs for the actuators
- Created various MATLAB scripts that analyzed signal frequency response and ensured proper signal generation

Systems Engineer Intern | Quartus Engineering | San Diego, CA *January 2022 – May 2022*
- Tested flamethrower robot to find its most efficient parameters by discussing an in-house experimental setup with customer to ensure monetary, timeline, and tangible goals were met
- Built and calibrated camera assemblies in a cleanroom by preparing future materials, adhering to a strict construction schedule, troubleshooting software issues as needed, and observing customer requests
- Costed and designed a customer's heavy lifter assembly and corresponding lift test tooling. Documented, assembled, organized, and cleanroom-tested the physical end product using an overhead crane

Project Experience

Spaceshot Rocketry Team *August 2021 – December 2022*
- Assembled and tested active roll control systems composed of rack-and-pinion servo motors
- Advanced embedded system that adjusts servos according to control algorithm and sensor feedback utilizing C++
- Validated new avionics stack performance onboard test rocket, which successfully guided rocket to 10,000 feet, deployed control flaps when necessary, and recorded flight telemetry for future analysis

NASA Micro-g Neutral Buoyancy Experiment Design Team *August 2019 – June 2020*
- Designed astronaut tool tolerant of operating with regolith dust that collects lunar rock samples
- Built and tested low-fidelity prototype to assess possible assembly schemes
- Composed and arranged sections of technical papers describing our team's tool and task plan
- Established test protocols according to NASA standards for validating tool effectiveness in the Neutral Buoyancy Lab